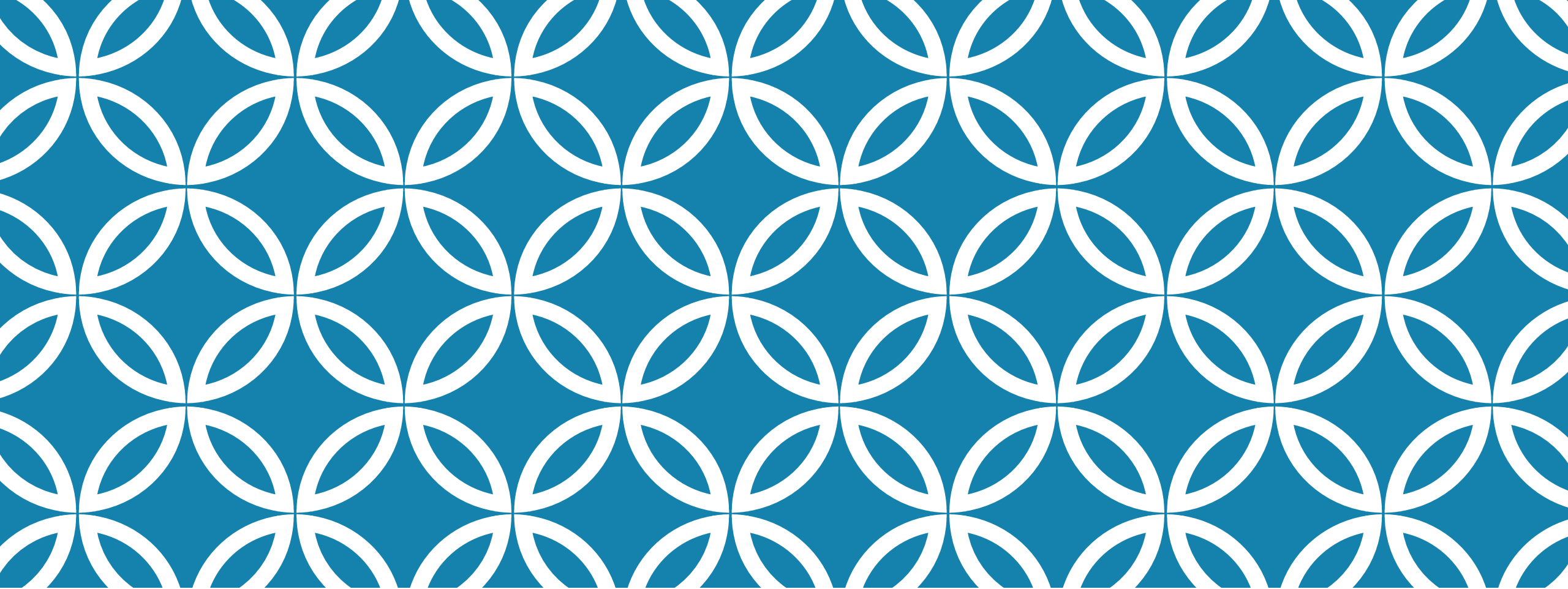




AI-DRIVEN SPARE PARTS INVENTORY OPTIMIZATION IN AUTOMOTIVE DEALERSHIPS

**SOLVING WORKSHOP BAY IDLENESS & WORKING CAPITAL LOCKUP IN AFTERSALES
OPERATIONS**

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Course Case Studies in AI for Business

THE AFTERSALES OPERATIONAL PROBLEMS & BOTTLENECK

How to Balance 10,000+ active SKUs in dealership networks ?

Why traditional inventory methods fail in workshop Service Centers ?

1. Missing Parts & delivery Delays

Blocked Service Bays: If a part isn't in stock, the car just sits there on the lift, and the technician has nothing to do

Customer dissatisfaction : This makes customers less happy, and if we order the part by air (DHL), it ends up costing a lot more than usual.

2. Overstocking & Dead stock

Locked Cash Flow: When we order extra parts "just in case," a lot of money ends up stuck in parts that just stay on the shelf for over a year.

Future Risk: And if parts like electronic modules sit too long without selling, once the model year changes we usually have to write them off as a loss.

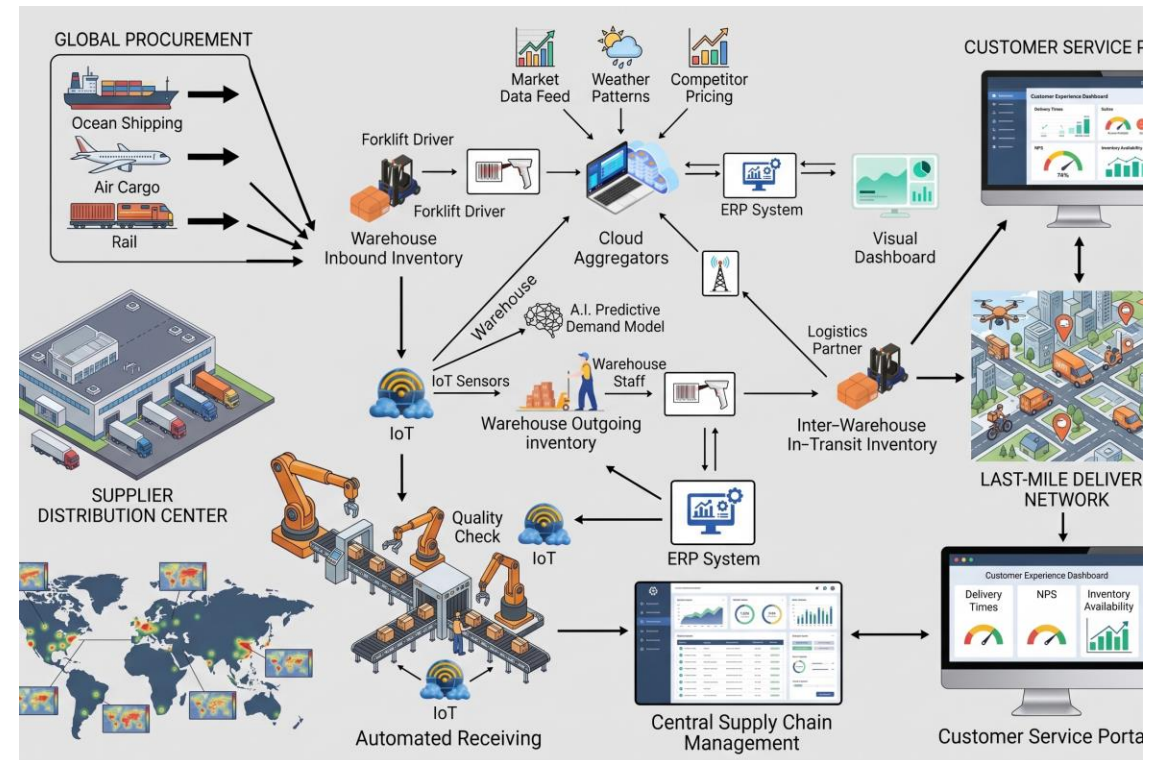
WHY OLD SYSTEMS FALL SHORT ?

Backward Metrics: The system reorders parts based only on the last 3 months of sales, without looking at upcoming bookings.

Seasonality prediction : It can't predict seasonal failures, like A/C compressors in summer or batteries in winter.

Ignores Mileage : It doesn't track the cars that are close to their 40k or 60k km service milestones.

Fixed Lead-Time changes : If the supplier's delivery time changes, the system doesn't adjust the safety stock.



AI-OPTIMIZED INVENTORY MANAGEMENT SYSTEMS ADVANTAGES



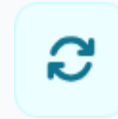
Appointments Integration

Connect to CRM booking Appointment, mileage prediction, and warranty records to forecast parts demand before repair orders are opened.



Dynamic Safety stock

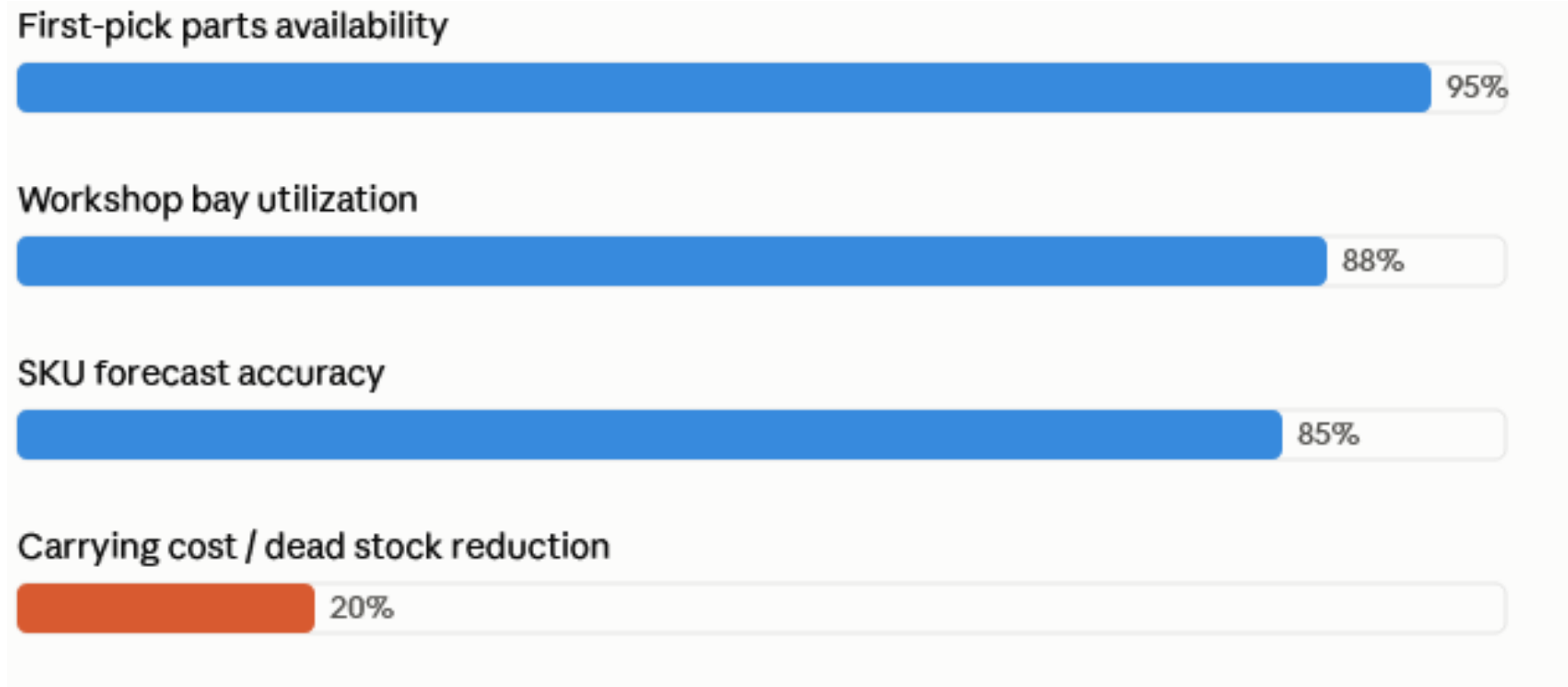
Instead of fixed Min/Max limits, the AI keeps adjusting safety stock on its own based on how reliable suppliers are and seasonal trends and any supply chain issues.



Automated recommendations

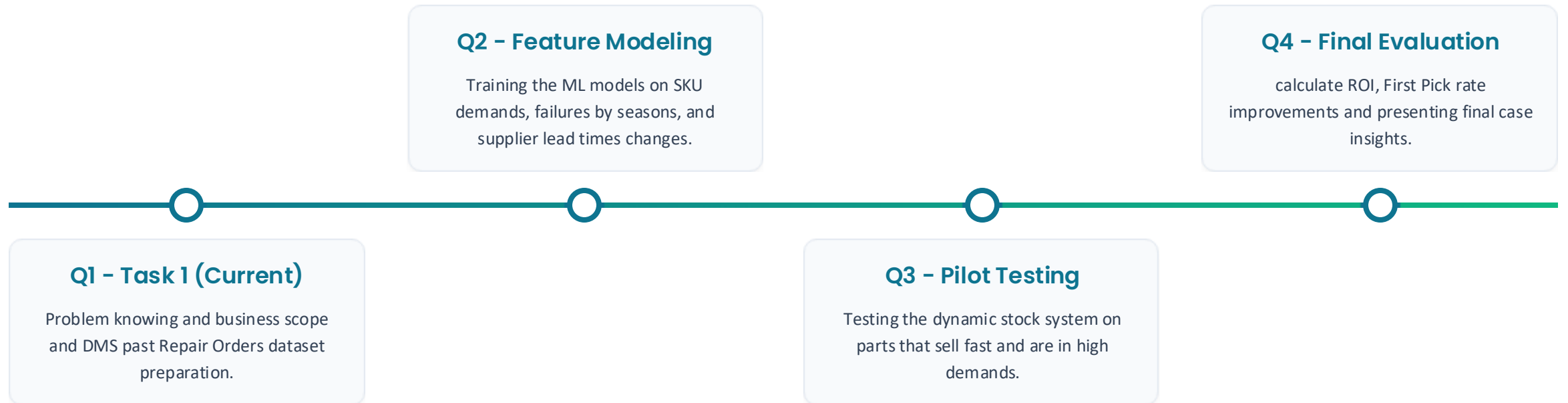
It create purchase order suggestions for the parts manager, who just checks them and gives the approval.

BUSINESS TARGET & IMPACT



When First-Pick availability is high the bays stay busy and more repair orders get closed each day and it also frees up to 20% of the cash that used to be tied up in inventory.

PROJECT SCOPE & ROADMAP



DATASET & ANALYTICS

THE DATASET WE'RE WORKING WITH

REAL SERVICE-CENTER RECORDS: WARRANTY CLAIMS, PAID REPAIRS, AND QUALITY RETURNS FOR SPARE PARTS



1,731

Indoor Paid

paid repair-order line items



833

Indoor Warranty

warranty claim line items



165

Outdoor Warranty & Quality

detailed claims w/ defect notes

Timeframe

~7 months of service-center activity

Vehicle models

Model 1, Model 2, Model 3, Model 4, Model 5, Model 6

Branches

SC 1 and SC 2

Combined scope

2,729 records across all three sources

WHAT EACH RECORD LOOKS LIKE

Core fields shared across the three data sources

Type	Claim category — Warranty Claim, Paid, Quality Claim, Returned Part
Model	Vehicle model the part belongs to (e.g., Model 1, Model 2)
Part Name	The specific spare part involved
Date	Month index of the claim (1–7 across the tracked period)
Branch	Service center — SC 1 or SC 2
Defect Description*	Free-text failure note (Outdoor sheet only)

** The Outdoor sheet also carries Customer Name, Dealer & Branch, an FR reference number, and a review Note (Accepted/etc.) — useful later for text-based failure categorization.*

BEFORE ANY MODELING: CLEANING THE DATA

KNOWN ISSUES IN THE RAW SHEETS THAT NEED HANDLING FIRST

1

Inconsistent text casing

Same model, three different spellings "MODEL 2", "Model 2", "model 2" all show up in the sheet.

2

Spacing variants

"Model 3" and "Model3" both show up too just a missing space, but enough to confuse a pivot table.

3

Sparse fields

Only the Outdoor sheet has FR NO and Note columns, and even there a lot of rows leave them blank.

4

Separate sources

Warranty and Paid data sit in separate sheets right now, so I'll need to join them on Model, Branch, and Date before I can analyze them together.

DATA PROCESSING PLAN

WORKING ENVIRONMENT: ANACONDA DISTRIBUTION, DEVELOPED IN A JUPYTER NOTEBOOK

01

Load

Read all six sheets into pandas DataFrames.

02

Clean

Standardize casing/spacing in Model and Part Name; drop duplicate rows.

03

Merge

Combine Warranty and Paid records into one unified claims table.

04

Engineer features

Add monthly aggregates per part, per model, and per branch.

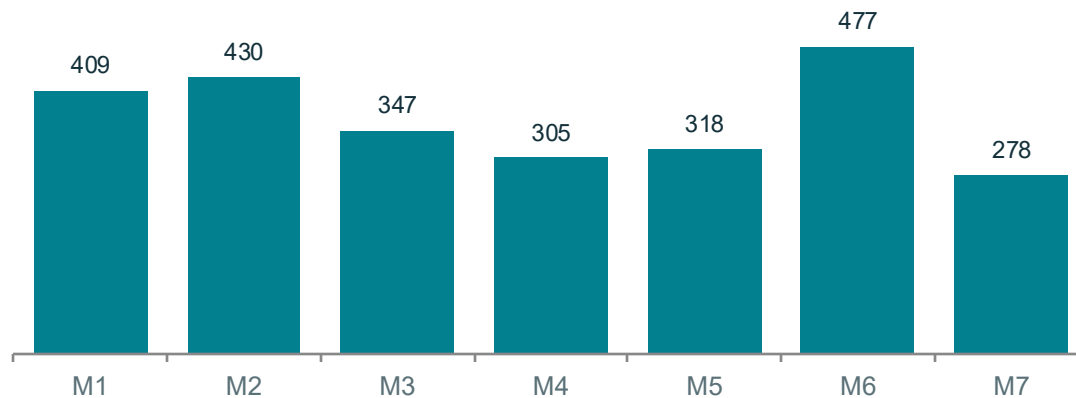
TOOLING

Anaconda distribution running a **Jupyter Notebook** as the working environment, with **pandas** for data handling.

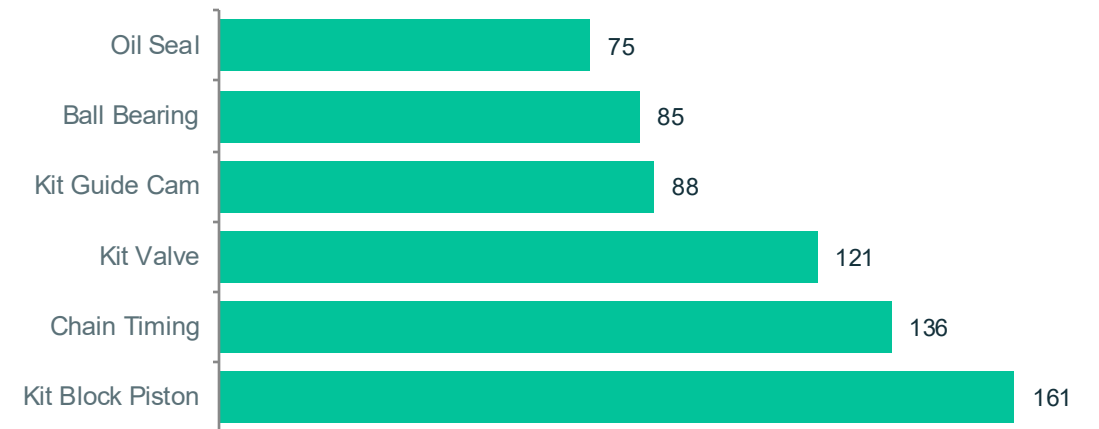
ANALYTICS DIRECTION

TWO PATTERNS FROM THE RAW DATA THAT WILL GUIDE THE FORECASTING MODEL

Monthly claim volume (warranty + paid)



Top replaced parts (warranty + paid)



Seasonality signal

There's a noticeable spike in month 6 - 403 paid claims. I want to check if that's seasonal or tied to some campaign that ran that month.

Concentration of demand

From all parts just 6 of them make up most of the claims. These are the first ones I'd set special stock rules for.

Next analysis step

Next I want to split that monthly trend by branch and model, to see if the spike is happening everywhere or just in one place.

AI FRAMEWORK: DESCRIPTIVE → PRESCRIPTIVE → SIMULATION

END-TO-END INVENTORY DECISION PIPELINE & VALIDATION METHODOLOGY

1. Service-Center Claims

Warranty, Paid & Quality records

2. Descriptive/Diagnostic Analytics

Monthly trend, top parts, model/branch mix

3. Predictive Forecast

Expanding-window per-part demand forecast

4. Prescriptive Policy

Dynamic reorder point & safety stock

1. Demand Forecasting

- Expanding-window average: uses all history up to the current month per part.
- Variability: standard deviation of that same history drives the safety margin.
- Recalculated every month, per part — not a single fixed number.

2. Dynamic Reorder Logic

- Reorder point = forecast + $1.65 \times \sigma$ ($\approx 95\%$ service-level factor).
- Order-up-to level = forecast + $2.65 \times \sigma$.
- Both recompute automatically as new months of data arrive.

3. Simulation Methodology

- Replays 7 real months, part by part, for the top 30 parts (67% of claim volume).
- Each policy starts from the same initial stock for a fair comparison.
- 210 part-months simulated per policy — no synthetic demand data used.

SIMULATION SETUP: CALIBRATION & PARAMETERS

DIRECTLY PARAMETERIZED USING OUR OWN ANONYMIZED CLAIMS DATA — NO SYNTHETIC INPUTS

Empirical Calibration Parameters

Directly measured from the anonymized service-center dataset:

Parts simulated	Top 30 (67% of claim volume)
Simulated period	7 months (real claims history)
Part-months simulated	210 per policy
Total demand units	1,719 claims
Safety factor (z)	1.65 (≈95% service level)
Forecast method	Expanding-window mean & std. dev.

Step 1: Baseline Policy

Fixed Min/Max per part — Min = 7-month average demand, Max = 2× average.
Reorders only after stock drops below Min.

Step 2: Dynamic Policy

Reorder point recalculated every month from all history available up to that point, plus a 1.65σ safety margin.

Step 3: Side-by-Side Replay

Both policies replay the exact same monthly demand, part by part, so only the reorder logic differs.

SCOPE NOTE

This simulation uses only fields present in our dataset (claim counts, model, branch, month). It does not use unit costs, lead times, or supplier data — those aren't in the current dataset, so no financial (\$) results are claimed at this stage.

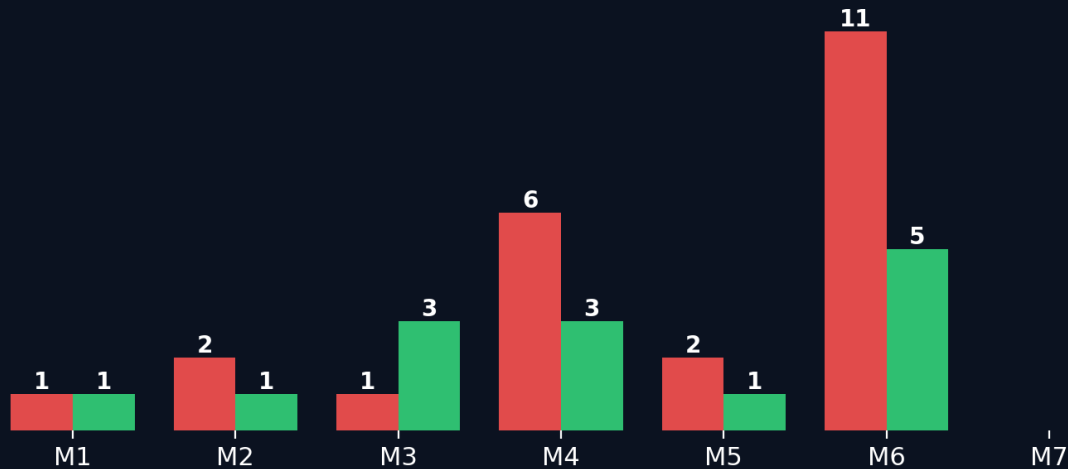
SIMULATION RESULTS

BASELINE VS AI-RECOMMENDED

SIMULATED OPERATIONAL PERFORMANCE ACROSS 210 PART-MONTHS OF REAL CLAIMS

Simulated Stock-out Events per Month (Top 30 Parts)

Baseline (Fixed Min/Max) AI-Recommended (Dynamic)



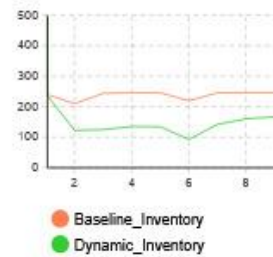
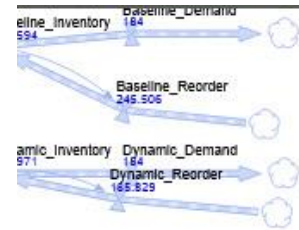
Metric	Baseline	AI-Recommended
Stock-out events	23	14 (−39.1%)
Service level	96.7%	97.8% (+1.1 pts)
Avg. stock held (units)	14.0	16.9 (+20.7%)
Reorder frequency	124	159 (+28.2%)

SIMULATION TAKEAWAY

The dynamic policy cuts simulated stock-outs by 39%, concentrated in Month 6 (the seasonal spike identified in Week 2) — but it holds about 21% more average stock and reorders more often. It's a real trade-off: fewer shortages in exchange for more capital tied up in inventory.

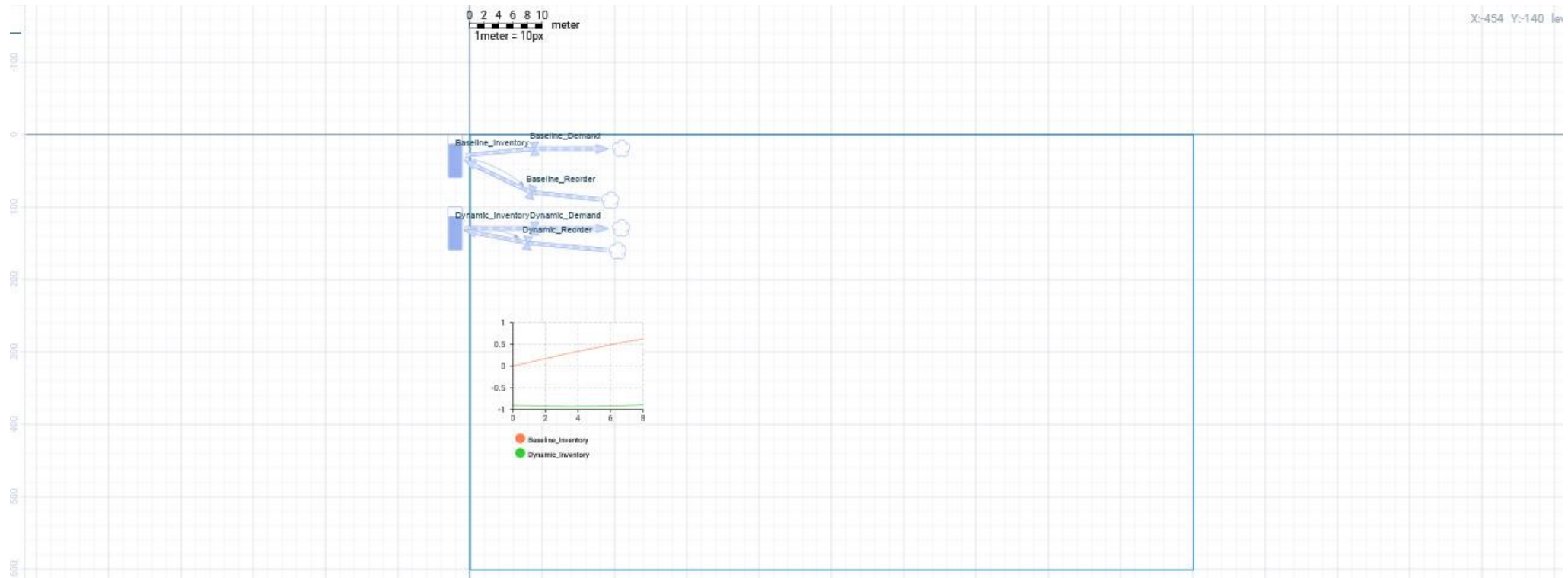
SIMULATION RESULTS

ANI LOGIC



SIMULATION RESULTS

ANY LOGIC



BUSINESS IMPACT & WHAT'S NEEDED NEXT

OPERATIONAL READ-OUT TODAY — FINANCIAL ROI REQUIRES ADDITIONAL DATA

What we can claim today

- 39% fewer simulated stock-outs on the top 30 parts
- Service level up from 96.7% to 97.8%
- Biggest gains land exactly in the Month-6 seasonal spike

The trade-off to weigh

- ~21% more average stock held to achieve that
- ~28% more (smaller, more frequent) reorders
- More manager touch-points, not necessarily more manager time

What we'd need for \$ ROI

- Unit cost per part (not in current dataset)
- Supplier lead times, to size real safety stock in days
- Holding cost rate, to price the extra stock trade-off

NEXT STEP

Request unit-cost and lead-time data from the dealership so the same simulation can be re-run to produce a defensible financial ROI, instead of an estimated one.

QUESTIONS & DISCUSSION

Thank you Dr. Ahmed & colleagues. I welcome your questions and feedback on this case idea.

Youssef Ashraf Ahmed | Case Studies in AI for Business